

inventory and audit processes.

Worldwide demand for smart cards and contactless ticketing solutions will continue to grow in every sector. Wireless identification of airport assets, like trolleys and roll-cages,

will help optimise the airport's asset management. Industries will be able to optimise and automate their production, supply chain management, component tracking and distribution. RFID tags will enable tracking

and tracing of goods from supplier to customers for logistics operations. This will increase visibility within the logistics chain for managing returnable transit items.

—Deepshikha Shukla

4 AI-POWERED DRONES COME TO THE RESCUE

A number of companies are manufacturing new-generation smart drones that use disruptive technologies to offer intelligent solutions in law enforcement, rescue and military operations, and provide their services for professional use in various countries

Ever since drones or unmanned aerial vehicles (UAVs) were first made, these have played an important role in several applications including wildlife monitoring, geographic mapping, agriculture, weather forecast, shipping and delivery, due to such advantages as portability, high efficiency and low operating costs. Many of these applications require cameras to be mounted on drones to capture images and videos.

When artificial intelligence (AI) is used in conjunction with drones, the latter can run on their own and offer a faster and more efficient response in critical real-time applications as well as make future predictions. AI-powered drones can perceive their surroundings, which is crucial to mapping areas, tracking objects and providing real-time analytical feedback to serve many purposes, especially when it comes to disaster relief and humanitarian aid operations.

In usual cases, the response force manually scans areas in their vehicles to get information about stranded people and the extent of destruction. By amplifying critical signals, it is possible to automatically send drones for aid and alerts to responders, too.

Being able to predict the most vulnerable targets in case of forthcoming natural disasters, whose numbers have significantly grown over the past decades, reduces the chances of severe damage to life and property. For instance, by having a more advanced solution that can work with the pre-existing weather forecasting and alerting system, Boston-based startup ClimaCell predicted the recent floods

in many parts of India like Kerala a few days in advance.

An end-to-end smart solution saves time by providing an integrated two-way approach to disaster management. Software for drones lets users keep a tab on the drones present in different locations and collaborate with each other. Hardware advancements make it possible for them to serve in such high-risk areas as wildfires and dangerous radiation zones, which was previously impossible. With proper communication and navigation transmission, drones can even reach such places as building interiors and underground facilities where ground-based rescue teams cannot, and help in locating missing survivors, potentially saving lives.

One such example is Eye in the Sky device created by students from Centre For Innovation, IIT Madras, which uses AI and computer vision to identify humans in disaster-hit locations, communicate with them and provide real-time updates to the relief task force. The team plans to work with other companies that can assist in conducting spot surveillance of disaster-struck areas.

By using several such drones, affected areas can be completely analysed within no time, thereby providing a channel for supplying necessities and safety aids to the trapped people. This potential has attracted the interests of many entrepreneurs, businesses and governments. Several manufacturers are now trying to integrate AI in UAVs.

Founded in 2014, American company Zipline builds and operates

drones that provide speedy delivery of medical products, primarily blood in emergencies. While the company's primary focus is Africa, government of Maharashtra recently announced that beginning in 2020, Zipline will be used to deliver emergency medicines in the state, outpacing conventional supply transport.

Besides Zipline, other such companies as Aerovironment, Shield AI, Applied Aeronautics and DroneSense are also manufacturing new-generation smart drones that use disruptive technologies to offer intelligent solutions in law enforcement, rescue and military operations, and provide their services for professional use in various countries. According to a report by Market Study Report, the military drone market is estimated to surpass the fleet of 18,000 units with a revenue of over thirteen billion dollars by 2024.

Exploring the possibility of helping human pilots in the Air Force for crucial missions, the US Air Force Research Laboratory is working on AI-powered prototypes of autonomous drones under its Skyborg programme. While the programme is still in its early stages, if successful, the algorithms could range from simple to complicated, and the launch is targeted by the end of 2023.

Although, armed drones have been used for a long time in the military but with their technological evolution, safety concerns associated with their use in warfare have increased. When utilised rightly to its entire potential, AI-powered drones have the power to achieve futuristic goals. **EFY**

—Ayushee Sharma